

Flow charts and operations



$$1 \quad \begin{array}{ccccc} & -7 & & \times 6 & \\ s & \rightarrow & s - 7 & \rightarrow & 6(s - 7) \end{array}$$

$$2 \quad \begin{array}{ccccc} & +3 & & \div 5 & \\ t & \rightarrow & t + 3 & \rightarrow & \frac{(t + 3)}{5} \end{array}$$

$$3 \quad \begin{array}{ccccc} \text{a} & & \times 5 & & +8 \\ y & \rightarrow & 5y & \rightarrow & 5y + 8 \end{array}$$

$$\text{b} \quad \begin{array}{ccccc} & \div 3 & & -7 & \\ k & \rightarrow & \frac{k}{3} & \rightarrow & \frac{k}{3} - 7 \end{array}$$

$$\text{c} \quad \begin{array}{ccccc} & +15 & & \times 8 & \\ p & \rightarrow & p + 15 & \rightarrow & 8(p + 15) \end{array}$$

$$\text{d} \quad \begin{array}{ccccc} & +5 & & \div 3 & \\ n & \rightarrow & n + 5 & \rightarrow & \frac{n + 5}{3} \end{array}$$

$$\text{e} \quad \begin{array}{ccccc} & \times 4 & & +3 & \\ x & \rightarrow & 4x & \rightarrow & 4x + 3 \end{array}$$

$$\text{f} \quad \begin{array}{ccccc} & \div 9 & & -7 & \\ j & \rightarrow & \frac{j}{9} & \rightarrow & \frac{j}{9} - 7 \end{array}$$

$$\text{g} \quad \begin{array}{ccccc} & +28 & & \times 3 & \\ p & \rightarrow & p + 28 & \rightarrow & 3(p + 28) \end{array}$$

$$\text{h} \quad \begin{array}{ccccc} & -6 & & \times 7 & \\ s & \rightarrow & s + 5 & \rightarrow & 7(s - 6) \end{array}$$

Operations in reverse order

$$4 \quad \begin{array}{ccccc} \text{a} & & -9 & & \div 6 \\ 6y + 9 & \rightarrow & 6y & \rightarrow & y \end{array}$$

$$\text{b} \quad \begin{array}{ccccc} & \div 5 & & -3 & \\ 5(q + 3) & \rightarrow & q + 3 & \rightarrow & q \end{array}$$

$$\text{c} \quad \begin{array}{ccccc} & \div 6 & & +9 & \\ 6(s - 9) & \rightarrow & s - 9 & \rightarrow & s \end{array}$$

$$\text{d} \quad \begin{array}{ccccc} & \times 5 & & -8 & \\ \frac{n + 8}{5} & \rightarrow & n + 8 & \rightarrow & n \end{array}$$

$$\text{e} \quad \begin{array}{ccccc} & +6 & & \times 5 & \\ \frac{j}{5} - 6 & \rightarrow & \frac{j}{5} & \rightarrow & j \end{array}$$

$$\text{f} \quad \begin{array}{ccccc} & -9 & & \div 8 & \\ 8y + 9 & \rightarrow & 8y & \rightarrow & y \end{array}$$

5

a $4(q+9) \xrightarrow{\div 4} q+9 \xrightarrow{-9} q$



b $5(r-7) \xrightarrow{\div 5} r-7 \xrightarrow{+7} r$

c $\frac{n+6}{9} \xrightarrow{\times 9} n+6 \xrightarrow{-6} n$

d $\frac{j}{2} - 7 \xrightarrow{+7} \frac{j}{2} \xrightarrow{\times 2} j$

Two-step equations

6

a Operation 1: Subtract 16
Operation 2: Divide by 4

b $24 \xrightarrow{\text{Operation 1}} 8 \xrightarrow{\text{Operation 2}} 2$

c $x = 2$

7

a Operation 1: Divide by 3
Operation 2: Subtract 11

b $57 \xrightarrow{\text{Operation 1}} 19 \xrightarrow{\text{Operation 2}} 8$

c $p = 8$

8

a Operation 1: Add 11
Operation 2: Multiply by 7

b $-7 \xrightarrow{\text{Operation 1}} 4 \xrightarrow{\text{Operation 2}} 28$

c $j = 28$

9

a i Subtract 6 from both sides, then divide both sides by 7.

ii $p = 5$

b i Add 5 to both sides, then divide both sides by 2.

ii $x = 8$

c i Divide both sides by 5, then add 15 to both sides.

ii $n = 22$

d i Divide both sides by 10, then subtract 10 from both sides.

ii $p = 2$



e

i Subtract 3 from both sides, then multiply both sides by 8.

ii $y = 16$

f

i Add 11 to both sides, then multiply both sides by 3.

ii $x = 60$

g

i Multiply both sides by 6, then subtract 22 from both sides.

ii $u = 2$

h

i Multiply both sides by 2, then add 10 to both sides.

ii $y = 20$